

PHILIPS

Efficia CM Series
Patient Monitors

Release C.00

English



Data Export Guide

Efficia CM Series Patient Monitors

Data Export Guide

C.00

English

PHILIPS

Notice

Proprietary Information

This document contains proprietary information, which is protected by copyright.

Copyright

Copyright © 2019 Koninklijke Philips N.V.
All Rights Reserved

Trademark Acknowledgments

Efficia is a registered trademark of Koninklijke Philips N.V. Other product names may be trademarks of their respective owners.

Manufacturer

Philips Medical Systems
3000 Minuteman Road
Andover, MA 01810
(978) 687-1501

Document Number
453564623581 Rev B

Warranty Disclaimer

The information contained in this document is subject to change without notice. Philips Medical Systems makes no warranty of any kind with regard to this material, including, but not limited to, the implied warranties or merchantability and fitness for a particular purpose. Philips Medical Systems shall not be liable for errors contained herein or for incidental or consequential damages in connection with the furnishing, performance, or use of this material.

Printing History

New editions of this document incorporate all material updated since the previous edition. Update packages may be issued between editions and contain replacement and additional pages to be merged by a revision date at the bottom of the page. Pages that are rearranged due to changes on a previous page are not considered revised.

The documentation printing date and part number indicate its current edition. The printing date changes when a new edition is printed. (Minor corrections and updates that are incorporated at reprint do not cause the date to change.) The document part number changes when extensive technical changes are incorporated.

Second Edition February 2019

Conventions

This section describes the conventions used in this guide.

Text Formatting

The following typographical conventions are used in this guide.

Typeface	Usage	Example
Bold	System keys	Press the Main Screen key.
Special bold	User interface text	Open the System Menu .
<i>Italic</i>	Variables, notes, document titles	<product name>-<hardware configuration>-<software version>.cfg <i>Efficia CM Series Patient Monitors Instructions for Use</i>

Decimal Points

Because the Efficia CM Series monitor uses a period (.) to indicate a decimal point in decimal numbers (for example, 10.0), all decimal numbers in this guide use a period as a decimal point. Commas are not use as decimal points.

Notes, Cautions, and Warnings

This guide uses the following conventions for Notes, Cautions, and Warnings.

Note — A **Note** calls attention to an important point in the text.

Caution A **Caution** calls attention to a condition or possible situation that could damage or destroy the product or the user's work.

Warning A **Warning** calls attention to a condition or possible situation that could cause injury to the user and/or patient.

1. Introduction

Overview	1-1
Data Export on an Efficia Central Network	1-1
HL7 Conformance	1-2
System Requirements	1-2
Intended Audience	1-2
Efficia CM Series Documentation	1-2

2. LAN/WLAN Data Export

LAN/WLAN Data Export Overview	2-1
Efficia Central Considerations	2-2
Flow of Events	2-2
Message Conventions	2-3
Notation Conventions	2-3
Block Format	2-4
Delimiter Characters	2-4
Segment Data Fields	2-4
Observation Reporting (ORU) Message Syntax	2-5
Message Header (MSH) Segment	2-7
Patient Identification (PID) Segment	2-9
Patient Visit (PV1) Segment	2-10
Common Order (ORC) Segment	2-11
Observation Request (OBR) Segment	2-11
Observation/Result (OBX) Segment	2-12
Acknowledgement (ACK) Message Syntax	2-17
Message Header (MSH) Segment	2-18
Message Acknowledgement (MSA) Segment	2-19
MDIL Coding	2-20
Parameters and Unit Sources	2-20
Alarms	2-26

3. Serial Data Export

Serial Data Export Overview	3-1
Exporting Parameters Displayed in Numeric Panes	3-1
Flow of Events	3-2
Message Conventions	3-2
Notation Conventions	3-3
Block Format	3-3
Delimiter Characters	3-4
Segment Data Fields	3-4
Query (QRY) Message Syntax	3-5
Message Header (MSH) Segment	3-5
Query Definition (QRD) Segment	3-7
Observation Reporting (ORF) Message Syntax	3-9
Message Header (MSH) Segment	3-10
Message Acknowledgement (MSA) Segment	3-12
Query Definition (QRD) Segment	3-13
Patient Identification (PID) Segment	3-14
Patient Visit (PV1) Segment	3-15
Common Order (ORC) Segment	3-16
Observation Request (OBR) Segment	3-16
Observation/Result (OBX) Segment	3-17

Acknowledgement (ACK) Message Syntax 3-20

Negative ACK Message Syntax. 3-21

 Message Header (MSH) Segment. 3-21

 Message Acknowledgement (MSA) Segment 3-23

Positive ACK Message Syntax. 3-24

 Message Header (MSH) Segment. 3-25

 Message Acknowledgement (MSA) Segment. 3-26

MDIL Coding 3-27

 Parameters and Unit Sources 3-27

4. Time Synchronization

Maintaining Synchronization on a LAN System 4-1

Introduction

This guide describes the syntax and structure of the Health Level Seven (HL7) messages used for patient records that are exported from Efficia® CM Series monitors.

Overview

The Efficia CM Series Patient Monitors can export patient data to a receiving system (HL7 Interface server) and then forward the data to an Electronic Health Records (EHR) system, or send the data directly to the EHR system.

The exported data includes:

- Vital signs measurements
- All patient identifiers except age
- High and low physiological alarms

The Efficia CM Series monitor can be configured to export patient data to a receiving system on your network using one of the following connections:

- A wired Local Area Network (LAN) connection
- A wireless Local Area Network (WLAN) connection
- An RS-232 serial connection using the optional USB to RS-232 serial interface adapter

For information on enabling the Data Export function and configuring the network connection, see the *Efficia CM Series Network Configuration Guide*.

Data Export on an Efficia Central Network

If the monitor is connected to an Efficia Central network, patient data is sent from the connected monitor to the Efficia Central and then exported from the Efficia Central to the receiving system on the hospital LAN network. For information about connecting a monitor to the Efficia Central network, enabling the Data Export function, and configuring the receiving system's network settings, see the *Efficia CM Series Central Installation and Configuration Guide*.

Note — *The Efficia CM monitor serial port cannot be used to connect to a Efficia Central. The serial port is for data export to an EHR server only.*

HL7 Conformance

The exported patient data is sent in messages that comply with the Health Level Seven (HL7) standard. This document describes the syntax and structure of the HL7 messages used for patient data exchanged between the Efficia CM Series monitor and the receiving system. For details about these messages, see Chapter 2, “LAN/WLAN Data Export” and Chapter 3, “Serial Data Export.”

Note — For a complete description of HL7 message syntax, see the *Health Level Seven Standard Version 2.4* available from Health Level Seven International (www.hl7.org).

System Requirements

The monitor and network must meet minimum requirements for LAN, WLAN, or serial data export. For information about the system requirements for LAN, WLAN, or serial data export, see the *Efficia CM Series Network Configuration Guide*.

Intended Audience

This guide is for individuals involved in planning, designing, and implementing an application that will receive the data exported from Efficia CM Series monitors. Readers should be familiar with Version 2.4 of the HL7 standard.

Efficia CM Series Documentation

Efficia CM Series documentation includes:

- *Efficia CM Series Patient Monitors Instructions for Use*: Provides information for day-to-day operation of the Efficia monitors. Also includes safety information, monitor specifications, and a list of compatible accessories.

Note — For information about purchasing additional copies of the printed *Efficia CM Series Patient Monitors Instructions for Use*, contact the Philips Customer Care Center.

- *Efficia CM10 and CM12 Quick Card* and *Efficia CM100, CM120, and CM150 Quick Card*: Provides brief descriptions of commonly used CM Series monitor functions.

The Efficia CM Series documentation CD contains PDF versions of the following documents:

- *Efficia CM Series Patient Monitors Installation and Configuration Guide*: Provides instructions for unpacking, installing, and connecting all hardware. Includes initial

testing and configuration procedures. Also includes instructions for returning the monitor.

- *Efficia CM Series Patient Monitors Service Guide*: Provides information about maintaining, troubleshooting, and performance verification testing of the monitor.
- *Efficia CM Series Patient Monitors Series Data Export Guide*: Provides detailed information about the syntax and structure of the HL7 messages that are exported from the monitors.
- *Efficia CM Series Patient Monitors Network Configuration Guide*: Provides instructions for configuring your monitor to connect to a network using a wired (LAN) connection, a wireless (WLAN) connection, or an RS-232 serial adapter.

LAN/WLAN Data Export

This chapter describes the flow of events and the conventions and syntax of the HL7 messages used to send data over a wired (LAN) or wireless (WLAN) network.

For information about configuring the LAN/WLAN network and the Data Export function, see the *Efficia CM Series Patient Monitors Installation and Configuration Guide*.

LAN/WLAN Data Export Overview

Use the **LAN** Data Export option to export patient records to a wired network; use the **LAN/WLAN** Data Export option to export patient records to a wired or wireless network. For wireless data export, an internal radio board provides the wireless connection between the monitor and the receiving system (server).

Data exported over a wired or wireless network can be encrypted. For information about enabling encryption, see the *Efficia CM Series Patient Monitors Installation and Configuration Guide*.

When the Efficia CM Series monitor is connected to a LAN or WLAN network and the Data Export function is enabled, the monitor exports data periodically and aperiodically.

The periodic export occurs at 60-second intervals, and if connected to the server includes:

- The most recent record in the trend database (saved records are white).

Aperiodic export includes NBP measurements. These measurements are sent at the time that they are complete. Also, alarms are exported as soon as they occur.

After the server sends a positive acknowledgment (ACK) message to the monitor, the exported records in the tabular trend display change from white to green.

Only data with a primary ID is exported. If a record does not contain a primary ID (**ID Unknown** appears in the patient pane on the monitor), the data is not exported.

If the network is unavailable and the Data Export function is enabled, the trend data remains in the tabular trend display at the monitor, but the data is not exported. When the network connection is re-established, the monitor resumes exporting any alarms that were not exported, and new trend data (the monitor does not resend old trend data).

If the server finds an error in the exported data, it returns a negative ACK message to the monitor. The rejected record in the tabular trend display changes from white to blue. If programmed at the server, the negative ACK message may contain an error message that indicates the reason the record was rejected.

The user cannot edit rejected records; the records remain blue in the tabular trend display.

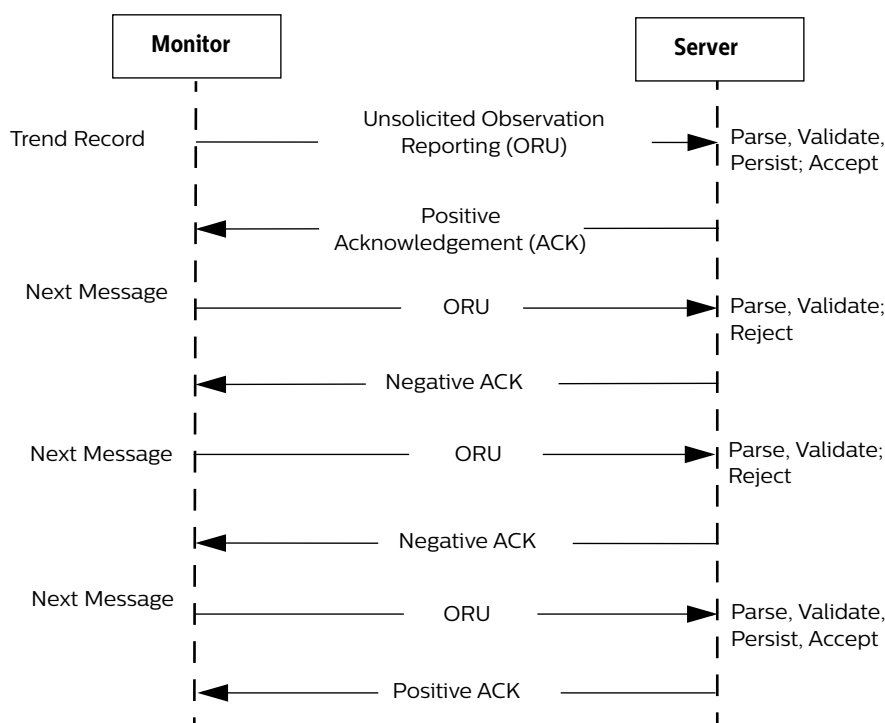
Efficia Central Considerations

If the monitor is connected to a Efficia Central network, note the following:

- Patient data is sent from the connected monitor to the Efficia Central and then exported from the Efficia Central to the receiving system (server) on the hospital network. For information about connecting a monitor to the Efficia Central network, enabling the Data Export function, and configuring the server's network settings, see the *Philips Efficia Central Installation and Configuration Guide*.
- The Efficia Central receives the positive or negative ACK message from the server and then returns the ACK message to the monitor.

Flow of Events

The following figure shows the flow of events when the sender (Monitor) sends a message to a receiving system (Server) over a wired (LAN) or wireless (WLAN) network.



1. The monitor sends an ORU message containing trend record results to the server.
2. The server receives the ORU message and attempts to process the message.
3. If the server successfully processes the message, it returns an ACK message with an AA (Application Accept) acknowledgment code to the monitor.
4. If the server is unable to process the message, it returns an ACK message with an AR (Application Reject) acknowledgment code to the monitor.

5. The monitor receives the ACK (AA) message, changes the trend record from white to green, and then starts to send the next ORU message (if any).
6. The monitor receives the ACK (AR) message, changes the trend record from white to blue, and then starts to send the next ORU message (if any).
7. If the server does not respond, the monitor retries the transmission. After three retries, the monitor stops trying and waits until the next interval (1 minute). The data on the monitor remains white until it exports.

Message Conventions

This section describes the format and conventions used in the HL7 messages used for LAN/WLAN export.

Message	Description
ORU	Unsolicited Observation Reporting. The Efficia CM Series monitor sends an ORU message containing patient data and vital signs measurements.
ACK	Acknowledgement. The receiving system (server) sends an ACK message to the Efficia CM Series monitor to acknowledge that the ORU message was received.

Note — For a complete description of HL7 message syntax, see the *Health Level Seven Standard Version 2.4* available from Health Level Seven International (www.hl7.org).

Notation Conventions

The HL7 messages use the following notation conventions:

- Single ASCII characters are enclosed in single quotes.
- Special characters or non-printing ASCII characters are enclosed in angle brackets, < >. Special characters are the Lower Layer Protocol (LLP) Start Block and End Block characters. Non-printing ASCII characters can be written as abbreviations; for example, ESC for the Escape character. They also may be written as hex values in the form 0xXX, where X is a hexadecimal digit. For example, in Standard ASCII, <ESC> is <0x1B>.
- The HL7 Minimal Lower Layer Protocol (MLLP) is used to mark message boundaries. Because TCP/IP is a byte-stream protocol and does not provide messaging boundaries, a wrapping protocol is required to recognize the start and end of HL7 messages. HL7, the standard for the upper-level protocol, is based on messages, but does not provide a mechanism to detect message termination.

Block Format

HL7 messages are enclosed by special characters to form a block. The format is as follows:

```
<SB>dddd<EB><CR>
```

where:

<SB> = Start Block character (1 byte). The ASCII vertical tab character <VT>, for example, <0x0B>. Do not confuse this with the ASCII characters SOH or STX.

dddd = Data (variable number of bytes). This is the HL7 data content of the block. The data can contain any ASCII characters and the carriage return character, <CR>, as a delimiter of individual message segments. UTF-8 encoding is used for languages that cannot use ASCII characters.

<EB> = End Block character (1 byte). The ASCII field separator character <FS>, for example, <0x1C>. Do not confuse this with the ASCII characters ETX or EOT.

<CR> = Carriage Return (1 byte). The ASCII carriage return character, for example, <0x0D>.

Delimiter Characters

The following characters are used to construct a message:

Character	Description
	Field Separator. Separates two adjacent data fields within a segment.
^	Component Separator. Separates adjacent components of data fields where allowed.
&	Subcomponent Separator. Separates adjacent subcomponents of data fields where allowed.
~	Repetition Separator. Separates multiple occurrences of a field where allowed.
\	Escape Character. Escape character for use with any field.

Segment Data Fields

Each message segment description contains the following data fields:

Character	Description
Seq	Sequence number of the field within the segment. Since an Efficia monitor does not use all possible fields in the HL7 standard, the sequence numbers are not always consecutive. This document shows only the fields that are included in the messages.
Field	Descriptive name for the field.
Description	Data field description.

Character	Description
DT	Data type of the field. Some common data types are: • ST = String • NM = Numeric • TS = Time stamp • CE = Coded element
Len	Maximum length of the field (for primitive types only).
Usage	Indicates whether the field is required or conditional in the segment. The abbreviations are: • R = Required element • RE = Required, but can be empty • C = Conditional – based on predicate evaluation • CE = Conditional, but can be empty
Min	Minimum number of occurrences.
Max	Maximum number of occurrences.
Table	Number of the HL7 table containing valid values for the field.
Sample Value	An example of the data field contents.

Observation Reporting (ORU) Message Syntax

The Unsolicited Observation Reporting (ORU) message that the monitor sends to the server contains patient data, including:

- Patient information
 - Medical Record Number (MRN) or other ID used as the patient identifier
 - First Name
 - Middle Name
 - Last Name
 - Gender
 - Age
 - Height
 - Weight
- Monitor Name
- Location ID (also called Bed ID)
- Transaction ID (also called Visit Number, Account Number, Finance Number (FIN))
- Operator ID
- Department
- Admit Time
- All measurements currently available on the monitor
- Any high or low alarms that have occurred since the data was last exported

Note — *The person configuring the monitor specifies which patient identifiers are required or optional, and selects the primary identifier. For more information about configuring patient IDs, see the Efficia CM Series Patient Monitors Installation and Configuration Guide.*

ORU Message Segments

The following table describes the segments within the ORU message:

Segment	Description	Usage	Min	Max
MSH	Message Header	R	1	1
PID	Patient Identification	R	1	1
PV1	Patient Visit	R	1	1
ORC	Common Order	R	1	1
OBR	Observation Request	R	1	1
OBX	Observation/Result	R	1	no max

Note — *Each message segment consists of a variable number of lines, each ending with a carriage return character. In the examples, long lines are broken for display purposes, and the carriage return character is indicated by <CR>.*

Examples:

The following example shows an ORU message sent by an Efficia CM Series monitor to the server. The message contains trend patient data.

```
MSH|^~\&|^EfficiaCM|||20140510124500||ORU^R01^ORU_R01|MONITORABC00000
0002D|P|
2.4|||||||2.16.840.1.113883.9.2.1<CR>
PID|||MRN1234||JONES^MARY^A<CR>
PV1|||^LOCATION34|||||||A|TRANSACTION1234<CR>
ORC|NW|||||||MONITORABC<CR>
OBR|||MONITOR||20120510124500||OPERATOR12<CR>
OBX|NM|0002-4bb8^SpO2^MDIL||98|0004-0220^%MDIL||||F<CR>
OBX|NM|0002-4a05^NBPs^MDIL||122|0004-
0f20^mmHg^MDIL||||F||20120510124430<CR>
OBX|NM|0002-4a06^NBPd^MDIL||85|0004-
0f20^mmHg^MDIL||||F||20120510124430<CR>
OBX|NM|0002-4a07^NBPm^MDIL||98|0004-0f20^mmHg^MDIL||||F<CR>
OBX|NM|0002-4182^HR^MDIL||80|0004-0aa0^bpm^MDIL||||F||||0002-
418a^ECG^MDIL<CR>
OBX|NM|0002-480a^Pulse^MDIL||85|0004-0aa0^bpm^MDIL||||F||||0002-
4bb8^SpO2^MDIL<CR>
OBX|NM|0002-50b0^etCO2^MDIL||5.7|0004-0f03^kPa^MDIL||||F<CR>
OBX|NM|0002-5012^awRR^MDIL||23|0004-0ae0^rpm^MDIL||||F<CR>
OBX|NM|0002-50ba^imCO2^MDIL||0.0|0004-0f03^kPa^MDIL||||F<CR>
```

```
OBX||NM|0007-f0c7^T1^MDIL||99.1|0004-1140^F^MDIL||||F<CR>
```

The following example shows an ORU message sent by the monitor to the server. The message contains a medium level alarm.

```
MSH|^~\&|^EfficiaCM||||20140510124500||ORU^R01^ORU_R01|MONITORABC00000
0002D|P|
2.4|||||||2.16.840.1.113883.9.2.1
PID||MRN1234||JONES^MARY^A<CR>
PV1||^LOCATION34|||||||A|TRANSACTION1234<CR>
ORC|NW|||||||MONITORABC<CR>
OBR||ALARM||20120513111801||OPERATOR12<CR>
OBX||TX|2^Yellow Alarm^MDIL-ALERT||SpO2 High, SpO2=98 %||||F<CR>
```

Message Header (MSH) Segment

Every ORU message begins with the MSH segment, which contains message envelope information, such as the source, destination, and some specifics of the message syntax. The MSH segment does not contain clinical information.

Example:

This MSH segment shows an ORU message being sent from the Efficia CM Series monitor on May 10, 2014, at 12:45:00 pm.

```
MSH|^~\&|^EfficiaCM||||20140510124500||ORU^R01^ORU_R01|MONITORABC00000
0002D|P|
2.4|||||||2.16.840.1.113883.9.2.1<CR>
```

The following table describes each field in the MSH segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Field Separator	The character that separates the segment ID (MSH) from the first data field (Encoding Characters) in the segment.	ST	1	R	1	1		
2	Encoding Characters	The four separator characters for the message in this order: component separator (^), repetition separator (~), escape character (\), subcomponent separator (&).	ST	4	R	1	1		^~\&
3	Sending Application	Generic identifier for the Efficia monitor.	HD	180	R	1	1	0361	^EfficiaCM

Message Conventions

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
3.2	universal ID	The identifier for all Efficia monitors.	ST	20	R	1	1		EfficiaCM
7	Date/Time of Message	The date and time that the monitor created the message.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDHHMMSS.	NM	20	R	1	1		20140510124500
9	Message Type	The server uses this field to know which data segments to recognize. Contains the message type, trigger event, and message structure ID.	CM_MSG	15	R	1	1	0076	ORU^R01^ORU_R01
9.1	message type	Unsolicited Observation Reporting (ORU) message.	ID	3	R	1	1	0076	ORU
9.2	trigger event	Unsolicited transmission of results: R01.	ID	3	R	1	1	0003	R01
9.3	message structure		ID	7	R	1	1	0354	ORU_R01
10	Message Control ID	The unique identifier for this message instance. The identifier is a combination of the 10-character device serial number and a 10-character message counter.	ST	20	R	1	1		MONITORABC000000002D
11	Processing ID	Indicates whether to process the message as part of a production (P), training (T), or debugging (D) system.	PT	3	R	1	1		

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
11.1	processing ID	All results from the monitor are production (P) messages.	ID	3	R	1	1	0103	P
12	Version ID	The HL7 version that the server matches to its own version to be sure the message will be interpreted correctly. HL7 is backwards compatible; any previous version is compatible with this specification.	VID	8	R	1	1	0104	
12.1	version ID	Version 2.4 is currently supported.	ID	3	R	1	1	0104	2.4
21	Conformance Statement ID	The HL7 registry object identifier (OID) corresponding to this message profile.	ID	30	R	1	1	0449	2.16.840.1.113883.9.2.1

Patient Identification (PID) Segment

The PID segment in the ORU message identifies the subject of the vital signs measurements.

Example:

This PID segment identifies the patient as Mary A. Jones. Her medical record number is MRN1234, her birth data is 1991/09/11.

```
PID|||MRN1234||JONES^MARY^A||19910911|F<CR>
```

The following table describes each field in the PID segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
3	Patient Identifier List	Contains the patient ID entered in the device.	CX	250	RE	0	1		
3.1	ID	Patient identifier. The medical record number (MRN) or another unique identifier used across the enterprise.	ST	20	RE	0	1		MRN1234

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
5	Patient Name	The patient's name.	XPN	45	RE	0	1		JONES^MARY^A
5.1	family name	The patient's family name.	FN	15	RE	0	1		
5.1.1	surname	The patient's last name.	ST	15	RE	0	1		JONES
5.2	given name	The patient's first name.	ST	15	RE	0	1		MARY
5.3	middle initial or name	The patient's middle name or initial.	ST	15	RE	0	1		A
7	Date of Birth	The patient's date of birth in the format YYYYMMDD	NM	8	RE	0	1		19910911
8	Gender	The patient's gender F – Female M – Male Blank – Unknown or not entered by the user	IS	2	RE	0	1		F

Patient Visit (PV1) Segment

The PV1 segment in the ORU message identifies the location ID, patient type and transaction (visit) ID.

Example:

This PV1 segment shows that the patient's bed is at LOCATION34. The patient is an adult and the visit ID is TRANSACTION1234.

```
PV1|||^LOCATION34|||||||A|TRANSACTION1234<CR>
```

The following table describes each field in the PV1 segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
3	Assigned Patient Location	The patient's assigned location.	PL	80	RE	0	1		^^LOCATION34
3.3	bed	Location ID for the bed.	IS	12	RE	0	1		LOCATION34
18	Patient Type	Identifies the patient type. The abbreviations are: • A = Adult • P = Pediatric • N = Neonatal	IS	2	R	1	1	0018	A

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
19	Visit Number	Unique number assigned to the patient visit.	CX	250	RE	0	1		
19.1	ID	Transaction or visit ID.	ST	20	RE	0	1		TRANSACTION1234
39	Department	Department to which patient is admitted	ST	20	RE	0	1		
44	Admit Time	Time when patient is admitted in the format YYYYMMDDHH MMSS	NM	14	RE	0	1		

Common Order (ORC) Segment

The ORC segment in the ORU message transmits the monitor name.

Example:

This ORC segment contains the monitor name MONITORABC.

```
ORC|NW|_|_|_|_|_|_|_|_|_|_|MONITORABC<CR>
```

The following table describes each field in the ORC segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Order Control	The function of the order segment. For example, NW = new order.	ID	2	R	1	1	O119	NW
18	Entering Device	Identifies the physical device (monitor) used to enter the vital signs measurements.	CE	250	R	1	1		
18.1	ID	Monitor name.	ST	10	R	1	1		MONITORABC

Observation Request (OBR) Segment

The OBR segment in the ORU message identifies the type of data being exported, the date and time that the measurements occurred, and the ID of the person taking the measurements. If the monitor is sending trend data, the data type in the OBR segment is MONITOR. If the monitor is sending an alarm, the data type in the OBR segment is ALARM.

Examples:

This OBR segment identifies the type of data as MONITOR. The measurements occurred at 12:45:00 pm on May 10, 2014 and were collected by OPERATOR12.

OBR|||MONITOR||20140510124500||OPERATOR12<CR>

This OBR segment identifies the type of data as ALARM. The alarm occurred at 12:45:00 pm on May 10, 2014 and were collected by OPERATOR12.

OBR||||ALARM||||20140510124500||||OPERATOR12<CR>

The following table describes each field in the OBR segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
4	Universal Service Identifier	The identifier code for the requested measurement.	CE	250	R	1	1		
4.1	identifier	Identifies the type of result. Valid values are: • MONITOR • ALARM	ST	20	R	1	1		MONITOR or ALARM
7	Observation Date/Time	Date and time of the measurement.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDHH MMSS.	NM	20	R	1	1		20140510124500
10	Collector Identifier	Identifies the person collecting the measurements.	XCN	20	RE	0	1		
10.1	ID number	The ID of the person collecting the measurements.	ST	12	RE	0	1		OPERATOR12

Observation/Result (OBX) Segment

Each OBX segment in the ORU message reports a single result or alert. There can be multiple OBX segments in an ORU message.

An OBX segment can report the following:

- Measurement results
- Alerts (physiological alarms)
- Weight and Height of the patient

Results

This section describes the OBX segment that reports trend data.

Examples:

These OBX segments contain the following trend data: SpO₂ (98%), Systolic NBP (122 mmHg), diastolic NBP (85 mmHg), mean NBP (98 mmHg), Heart Rate (80 bpm), Pulse Rate (85 bpm) derived from SpO₂, end tidal CO₂ (5.7 kPa), airway respiration rate (23 rpm), inspired minimum CO₂ (0.0 kPa); and temperature (99.1°F). All of the results are final.

```

OBX||NM|0002-4bb8^SpO2^MDIL||98|0004-0220^%^MDIL||||F<CR>
OBX||NM|0002-4a05^NBPs^MDIL||122|0004-
0f20^mmHg^MDIL||||F|||20140510124430<CR>
OBX||NM|0002-4a06^NBPd^MDIL||85|0004-
0f20^mmHg^MDIL||||F|||20140510124430<CR>
OBX||NM|0002-4a07^NBPm^MDIL||98|0004-
0f20^mmHg^MDIL||||F|||20140510124430<CR>
OBX||NM|0002-4182^HR^MDIL||80|0004-0aa0^bpm^MDIL||||F|||||0002-
418a^ECG^MDIL<CR>
OBX||NM|0002-480a^Pulse^MDIL||85|0004-0aa0^bpm^MDIL||||F|||||0002-
4bb8^SpO2^MDIL<CR>
OBX||NM|0002-50b0^etCO2^MDIL||5.7|0004-0f03^kPa^MDIL||||F<CR>
OBX||NM|0002-5012^awRR^MDIL||23|0004-0ae0^rpm^MDIL||||F<CR>
OBX||NM|0002-50ba^imCO2^MDIL||0.0|0004-0f03^kPa^MDIL||||F<CR>
OBX||NM|0002-f0c7^T1^MDIL||99.1|0004-1140^F^MDIL||||F<CR>

```

In this OBX segment, the **X** in the Observation Result Status field (OBX-11) indicates an invalid SpO₂ measurement:

```

OBX||NM|0002-4bb8^SpO2^MDIL|||0004-0220^%^MDIL||||X<CR>

```

The following table describes the fields in the OBX segment that reports results.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
2	Value Type	Format of the measurement in the Observation Value field. The format is NM (numeric).	ID	2	R	1	1	0125	NM
3	Observation Identifier	Contains a unique identifier for the measurement. The format is: <identifier>^ <text>^ <name of coding system>	CE	250	R	1	1		0002-4182^HR^MDIL
3.1	identifier	Code that uniquely identifies the measurement. The identifier is an MDIL code. For more information, see “MDIL Coding” on page 2-20.	ST	20	R	1	1		0002-4182
3.2	text	Name or description of the coded measurement.	ST	20	R	1	1		HR

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
3.3	name of coding system	For measurements, the coding system is MDIL. For more information, see “MDIL Coding” on page 2-20.	IS	20	RE	0	1	0396	MDIL
5	Observation Value	Contains the measurement value recorded on the monitor.	NM	100	RE	0	1		80
6	Units	Contains the units of measurement for the Observation Value. The format is: <identifier>^ <unit of measure>^ <name of coding system>	CE	250	R	1	1		0004-0aa0^bpm^MDIL
6.1	identifier	Code that uniquely identifies the unit of measurement. The identifier is an MDIL code. For more information, see “MDIL Coding” on page 2-20.	ST	20	R	1	1		0004-0aa0
6.2	text	The name or description of the unit of measure.	ST	20	R	1	1		bpm
6.3	name of coding system	For measurements, the coding system is MDIL. For more information, see “MDIL Coding” on page 2-20.	IS	20	RE	0	1	0396	MDIL

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
8	Abnormal Flags	Indicates the normalcy status of the result. Valid values are: • Empty = Normal • L = Below normal • H = Above normal • A = Other abnormal	ID	1	RE	0	1	0078	H
11	Observation Result Status	Contains the result status. Valid values are: • F = Final results • X = Results cannot be obtained	ID	1	R	1	1	0085	F
14	Observation Date/Time	Timestamp of the following measurements: • NBP • Pulse Rate derived from NBP Timestamp is in the format YYYYMMDDHHMMSS	NM	20	RE	0	1		20140510123500
17	Observation Method	The method used to obtain the result.	CE	250	RE	0	1		0002-4bb8^SpO2^MDIL
17.1	identifier	The code that corresponds to the Heart Rate source, Pulse Rate source, or temperature method. By default, these measurements use MDIL codes.	ST	20	RE	0	1		0002-4bb8
17.2	text	Text describing the measurement method. Valid methods are: • Pulse Rate source: SpO2 or NBP	ST	20	RE	0	1		SpO2

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
17.3	name of coding system	For Heart Rate source, Pulse Rate source, and temperature method, the default coding system is MDIL. For more information, see “MDIL Coding” on page 2-20.	IS	20	RE	0	1	0396	MDIL

Alarms

This section describes the OBX segment that reports alarms.

Example:

This OBX segment contains a medium-level physiological alarm.

```
OBX||TX|2^Yellow Alarm^MDIL-ALERT||HR Low, HR=55 /min|||||F<CR>
```

The following table describes the fields in the OBX segment that reports alarms.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
2	Value Type	Format of the alarm in the Observation Value field. The format is TX (text).	ID	2	R	1	1	0125	TX
3	Observation Identifier	Contains a unique identifier for the alarm. The format is: <identifier>^ <text>^ <name of coding system>	CE	250	R	1	1		2^Yellow Alarm^MDIL-ALERT
3.1	identifier	Code that uniquely identifies the alarm. For alarms, the identifier is an MDIL-ALERT code. For more information, see “MDIL Coding” on page 2-20.	ST	20	R	1	1		2
3.2	text	Description of the alarm priority.	ST	20	R	1	1		Yellow Alarm

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
3.3	name of coding system	For alarms, the coding system is MDIL-ALERT. For more information, see “MDIL Coding” on page 2-20.	IS	20	RE	0	1	0396	MDIL-ALERT
5	Observation Value	Contains the alarm message text.	TX	100	RE	0	1		HR Low, HR=55 /min
11	Observation Result Status	Contains the alarm status. Valid value is F (Final results).	ID	1	R	1	1	0085	F

Acknowledgement (ACK) Message Syntax

The Acknowledgement (ACK) message is sent by the server back to the Efficia CM Series monitor to acknowledge that the message was received and validated.

ACK Message Segments

The following table describes the segments within the ACK message.

Segment	Description	Usage	Min	Max
MSH	Message Header	R	1	1
MSA	Message Acknowledgement	R	1	1

The following syntax information represents the profile for the server. Any message segment that is not supported is ignored. RE elements (*required, but can be empty*) are used, if valid, but are ignored otherwise.

Note — Each message segment consists of a variable number of lines, each ending with a carriage return character. In the examples, long lines are broken for display purposes, and the carriage return character is indicated by <CR>.

Example:

The following example shows an ACK message sent from the server back to the monitor.

```
MSH|^~\&|||||201405010124500||ACK^^ACK_ALL|EXAMPLE123|P|2.4<CR>
MSA|AA|MONITORABC000000002D<CR>
```

Message Header (MSH) Segment

The MSH segment is used primarily as a message envelope.

Example:

This MSH segment shows an ACK message being sent back to the monitor on May 10, 2014 at 12:45:00 pm.

```
MSH|^~\&|||||20140510124500||ACK^^ACK_ALL|EXAMPLE123|P|2.4<CR>
```

The following table describes each field in the MSH segment .

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Field Separator	The character that separates the segment ID (MSH) from the first data field (Encoding Characters) in the segment.	ST	1	R	1	1		
2	Encoding Characters	The four separator characters for the message in this order: component separator (^), repetition separator (~), escape character (\), subcomponent separator (&).	ST	4	R	1	1		^~\&
7	Date/Time of Message	The date and time that the monitor created the message.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDHH MMSS	NM	20	R	1	1		20140510124500
9	Message Type	The server uses this field to know which data segments to recognize. Contains the message type, trigger event, and message structure ID.	CM_MSG	1	R	1	1	0076	ACK^^ACK_ALL
9.1	message type	Acknowledgment (ACK) message.	ID	3	R	1	1	0076	ACK
9.3	message structure		ID	7	R	1	1	0354	ACK_ALL

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
10	Message Control ID	The unique identifier for this message instance.	ST	20	R	1	1		EXAMPLE123
11	Processing ID	Indicates whether to process the message as part of a production (P), training (T), or debugging (D) system.	PT	3	R	1	1		
11.1	processing ID	All results from the monitor are Production (P) messages.	ID	3	R	1	1	0103	P
12	Version ID	The HL7 version that the server matches to its own version to be sure the message will be interpreted correctly. HL7 is backwards compatible; any previous version is compatible with this specification.	VID	8	R	1	1	0104	
12.1	version ID	Version 2.4 is currently supported.	ID	3	R	1	1	0104	2.4

Message Acknowledgement (MSA) Segment

The MSA segment in an ACK message contains the response code and the initiating message identifier.

Examples:

This MSA segment shows that the server successfully persisted the ORU message.

```
MSA|AA|MONITORABC000000002D<CR>
```

This MSA segment shows that the receiving system found an error in the ORU message. The receiving system returns a text message **Incorrect Patient ID** to the monitor.

```
MSA|AE|MONITORABC000000002D|Incorrect Patient ID<CR>
```

The following table describes each field in the MSA segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Acknowledgment Code	Contains the acknowledgment code. The following codes are supported: • AA = Accept • AR = Reject • AE = Error	ID	2	R	1	1	0008	AE
2	Message Control ID	Contains the message control ID of the initiating message (ORU R01).	ST	20	R	1	1		MONITORABC00000002D
3	Text Message	Contains a text message indicating the reason for the error.	ST	80	RE	0	1		Incorrect Patient ID

MDIL Coding

The Efficia CM Series monitors use numeric codes from the Medical Device Interface Language (MDIL) standard to uniquely identify parameters, unit sources, and alarms.

Parameters and Unit Sources

To identify parameters and unit sources, the MDIL coding system uses identifiers of the following format:

<Partition>-<TermCode>

The values for the Partition and TermCode are hexadecimal values between 0x0000 and 0xffff, which are separated in the output with a “-” character.

Partition may be one of the following values:

- 0x0002 – SCADA partition
- 0x0401 – EMFC partition
- 0x0004 – Dimension partition (for units)

Note — *The label strings are provided in English only. The strings are not translated into other languages. Philips recommends that for parameter identification, use the numeric parameter identifiers instead of parameter labels.*

The following table lists the MDIL codes for Efficia CM Series monitor measurements. Only those parameters that are available on your monitor are exported.

MDIL Code	Parameter Label	Description
0002-4bb8	SpO2	Oxygen saturation of arterial blood
0002-4a05	NBPs	Non-invasive blood pressure (systolic)
0002-4a06	NBPd	Non-invasive blood pressure (diastolic)
0002-4a07	NBPM	Non-invasive blood pressure (mean)
0002-4182	HR	Heart Rate
0002-480a	Pulse	Pulse Rate
0002-4a15	ABPs	Arterial blood pressure (systolic)
0002-4a16	ABPd	Arterial blood pressure (diastolic)
0002-4a17	ABPM	Arterial blood pressure (MAP)
0002-4a1d	PAPs	Pulmonary artery pressure (systolic)
0002-4a1e	PAPd	Pulmonary artery pressure (diastolic)
0002-4a1f	PAPM	Pulmonary artery pressure (MAP)
0002-4a47	CVPm	Central venous pressure (mean)
0002-50b0	etCO2	End tidal CO ₂ concentration
0002-50ba	imCO2	Inspired minimum CO ₂
0002-5012	awRR	Airway respiration rate
0002-5000	Resp	ECG Respiration
0002-0301	ST-I	ST of ECG Lead I
0002-0302	ST-II	ST of ECG Lead II
0002-033d	ST-III	ST of ECG Lead III
0002-033e	ST-aVR	ST of ECG Lead aVR
0002-033f	ST-aVL	ST of ECG Lead aVL
0002-0340	ST-aVF	ST of ECG Lead aVF
0002-0343	ST-V	ST of ECG Lead V
0002-034b	ST-MCL	ST of ECG Lead MCL
0002-0303	ST-V1	ST of ECG Lead V1
0002-0304	ST-V2	ST of ECG Lead V2
0002-0305	ST-V3	ST of ECG Lead V3
0002-0306	ST-V4	ST of ECG Lead V4
0002-0307	ST-V5	ST of ECG Lead V5

MDIL Code	Parameter Label	Description
0002-0308	ST-V6	ST of ECG Lead V6
0002-4261	PVC	Premature ventricular contraction
0002-fb62	SpCO	Carboxyhemoglobin saturation
0002-fb67	SpOC	Oxygen content
0002-fbcf	SpHb	Total hemoglobin
0002-fb5f	RRa	Acoustic respiration rate
0002-4bb0	Perf	Perfusion index
0002-4a11	ARTs	Arterial blood pressure (systolic)
0002-4a12	ARTd	Arterial blood pressure (diastolic)
0002-4a13	ARTm	Arterial blood pressure (mean)
0002-4a0d	Aos	Aortic pressure (systolic)
0002-4a0e	Aod	Aortic pressure (diastolic)
0002-4a0f	Aom	Aortic pressure (mean)
0002-4a01	Ps	Non-specific pressure (systolic)
0002-4a02	Pd	Non-specific pressure (diastolic)
0002-4a03	Pm	Non-specific pressure (mean)
0002-4a24	PAWP	Pulmonary artery wedge pressure
0002-4a29	UAPs	Umbilical arterial pressure (systolic)
0002-4a2a	UAPd	Umbilical arterial pressure (diastolic)
0002-4a2b	UAPm	Umbilical arterial pressure (mean)
0002-5809	ICPs	Intracranial pressure (systolic)
0002-580a	ICPd	Intracranial pressure (diastolic)
0002-580b	ICPm	Intracranial pressure (mean)
0002-4a31	LAPs	Left atrial pressure (systolic)
0002-4a32	LAPd	Left atrial pressure (diastolic)
0002-4a33	LAPm	Left atrial pressure (mean)
0002-4a35	RAPs	Right atrial pressure (systolic)
0002-4a36	RAPd	Right atrial pressure (diastolic)
0002-4a37	RAPm	Right atrial pressure (mean)
0002-4a49	UVPs	Umbilical venous pressure (systolic)
0002-4a4a	UVPd	Umbilical venous pressure (diastolic)
0002-4a4b	UVPm	Umbilical venous pressure (mean)

MDIL Code	Parameter Label	Description
0002-f0c7	T1	Temperature #1
0002-f0c8	T2	Temperature #2
0002-f0c9	T3	Temperature #3
0002-f0ca	T4	Temperature #4
0002-4b04	C.O.	Cardiac output
0002-e014	Tblood	Blood temperature
0002-f093	Weight	Patient weight
0002-f094	Height	Patient height

The following table lists the MDIL codes for the measurements of the plug-in modules connected to the Efficia Rack. Only those parameters that are available on your monitor are exported.

MDIL Code	Parameter Label	Description
0002-f04e	BIS	Bispectral Index
0002-593c	EMG	Electromyography
0002-5988	SEF	Spectral Edge Frequency
0002-f04d	SQI	BIS Signal Quality Index
0002-f04a	SR	EEG Suppression Ratio
0002-59b8	TP	Total Power
0002-59ac	BC	Burst Count
0002-f04e	BIS L	Left Bispectral Index
0002-593c	EMG L	Left Electromyography
0002-5988	SEF L	Left Spectral Edge Frequency
0002-f04d	SQI L	Left BIS Signal Quality Index
0002-f04a	SR L	Left EEG Suppression Ratio
0002-59b8	TP L	Left Total Power
0002-59ac	BC L	Left Burst Count
0002-f04e	BIS R	Right Bispectral Index
0002-593c	EMG R	Right Electromyography
0002-5988	SEF R	Right Spectral Edge Frequency
0002-f04d	SQI R	Right BIS Signal Quality Index
0002-f04a	SR R	Right EEG Suppression Ratio
0002-59b8	TP R	Right Total Power

MDIL Code	Parameter Label	Description
0002-59ac	BC R	Right Burst Count
0002-538c	etAGT	Gas Analyzer Agent - End Tidal
0002-5390	inAGT	Gas Analyzer Agent - inspired
0002-f81e	etAGTs	End Tidal Anesthetic Agent - secondary agent
0002-f81f	inAGTs	Inspired Gas Analyzer Agent - secondary agent
0002-522c	etN2O	Gas Analyzer End-Tidal Nitrous Oxide
0002-5280	inN2O	Gas Analyzer Nitrous Oxide - inspired
0002-5378	etO2	End-Tidal Oxygen Concentration
0002-5284	inO2	Oxygen Concentration - inspired
0002-5214	etDES	Desflurane Concentration - End Tidal
0002-5218	etENF	Enflurane Concentration - End Tidal
0002-521c	etHAL	Halothane Concentration - End Tidal
0002-5220	etSEV	Gas Analyzer Sevoflurane - End Tidal
0002-5224	etISO	Gas Analyzer Isoflurane - End Tidal
0002-5268	inDES	Desflurane Concentration - inspired
0002-526c	inENF	Enflurane Concentration - inspired
0002-5270	inHAL	Halothane Concentration - inspired
0002-5274	inSEV	Sevoflurane Concentration - inspired
0002-5278	inISO	Gas Analyzer Isoflurane - inspired
0402-f921	sVMode	Setting: Ventilation mode
0002-5012	awRR	Airway Respiration Rate
0002-514c	MVexp	Expired Minute Volume
0002-5109	PIP	Positive Inspiratory Pressure
0002-510b	MnAwP	Mean Airway Pressure
0002-50dd	PIF	Peak Inspiratory Flow
0002-f8a1	ExpTi	Expiratory Time
0002-513c	TV	Tidal Volume
0002-51a8	PEEP	Positive End-Expiratory Pressure
0002-f8a3	InsTi	Spontaneous Inspiration Time
0002-508c	Cdyn	Dynamic Lung Compliance
0002-7498	FiO2	Fractional Inspired Oxygen
0002-f819	RSBI	Rapid Shallow Breathing Index

MDIL Code	Parameter Label	Description
0002-50e0	VCO2	CO2 Production
0002-fa06	WOBv	Work of Breathing
0002-50b0	etCO2	End-Tidal CO2 Concentration
0002-50e8	Pplat	Plateau Pressure
0002-51cc	MMV	Mandatory Minute Volume
0002-f091	SpMV	Spontaneous Minute Volume
0002-f816	ETC	Time constant
0002-f815	SpAWRR	Spontaneous Airway Respiration Rate
0002-f897	TOFrat	Train of Four Ratio
0002-f8ab	TOFcnt	Train of Four (TOF) count - number of TOF responses
0002-f8ac	Twitch	Twitch height of the 1 Hz/0.1 Hz stimulation response
0002-f86b	DBSrat	Double Burst Ratio
0002-f88b	PTC	Post Tetanic Count stimulation

The following table lists the MDIL codes for units of measurement.

MDIL Code	Parameter Label	Description
0002-f099	MAC	Minimum Alveolar Concentration
0004-0220	%	Percent
0004-0f20	mmHg	Millimeters of mercury
0004-0f03	kPa	Kilopascals
0004-0aa0	bpm	Beats per minute
0004-17a0	C	Celsius
0004-1140	F	Fahrenheit
0004-0ae0	rpm	Respirations per minute
0004-0000	No label	No unit of measure
0004-09e0	/min	per minute
0004-0512	mm	Millimeter
0004-0f40	cmH2O	Centimeter of water
0004-0892	msec	Millisecond
0004-0840	g/dl	Grams per deciliter
0004-1272	mmol/l	Millimoles per liter
0004-1912	ml/dl	Milliliters per deciliter

MDIL Code	Parameter Label	Description
0004-0c00	l/min	Liters per minute
0004-06c3	kg	Kilogram
0004-06e0	lb	Pound
0004-0511	cm	Centimeter
0004-0560	in	Inch

The following table lists MDIL codes for Pulse Rate source.:

MDIL Code		Description
0002-4a07	NBPm	Mean NBP Pulse Rate source
0002-4bb8	SpO2	SpO ₂ Pulse Rate source

Alarms

The following MDIL-ALERT codes identify the severity of an alarm on the Efficia CM Series monitor:

MDIL-ALERT Code	Description
6 Soft Inop	Low priority (technical) alarm on the monitor.
2 Yellow Alarm	Medium priority alarm on the monitor.
1 Red Alarm	High priority alarm on the monitor.

Serial Data Export

This chapter describes the flow of events and the conventions and syntax of the HL7 messages used to send data to a receiving application using the serial interface adapter.

For information about configuring the network and the Data Export function, see the *Efficia CM Series Installation and Configuration Guide*.

Note — *The Efficia CM Series monitor serial port cannot be used to connect to a Efficia Central. The serial port is for data export to an EHR server only.*

Serial Data Export Overview

Use the **Serial** Data Export option to export all measurements results.

Exporting Parameters Displayed in Numeric Panes

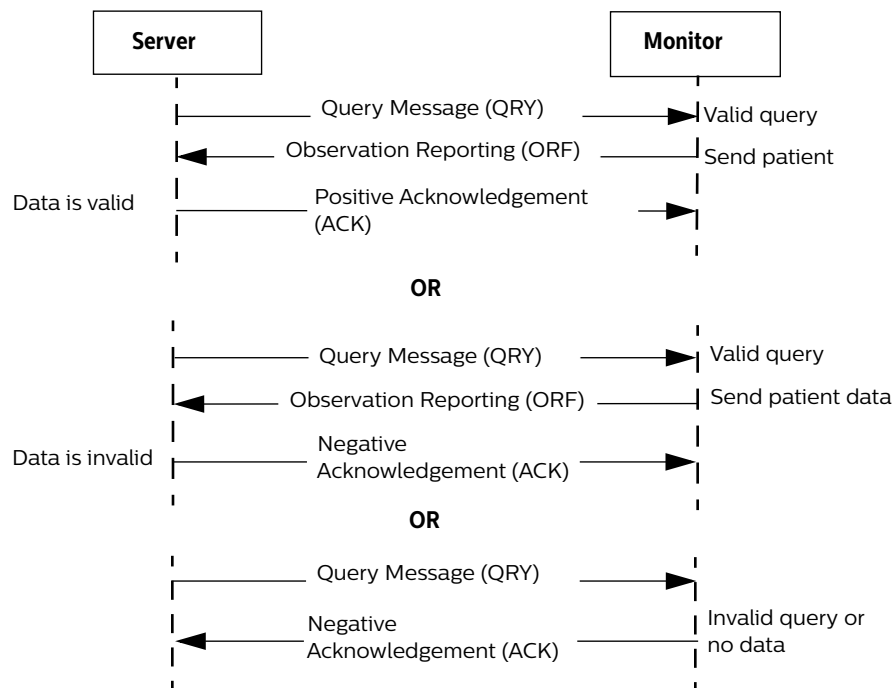
The monitor displays measurement data for the current patient in the numeric panes. The receiving application sends a Query (QRY) message to request all parameters. The QRY message contains STA (Status) in the What subject filter (QRD-9).

When the receiving application receives all of the data, it sends an Acknowledgement (ACK) message to the monitor.

When the monitor receives the QRY message, it sends all patient identifiers (or **ID Unknown**), valid and invalid measurements, blank results, and measurements that exceed high and low alarm limits. If the monitor determines that the QRY is invalid, it returns a negative ACK message to the receiving application.

Flow of Events

The following figure shows the serial data export flow of events.



- Messages are processed one by one.
- Every QRY message from the server generates exactly one ORF message or Negative ACK message from the monitor.

Message Conventions

This section describes the format and conventions used in the HL7 messages used for serial data export.

Message	Description
QRY	Query. The receiving system (server) sends a QRY message to the Efficia CM Series monitor requesting patient data.
ORF	Observation Reporting. The monitor sends an ORF message in reply to the QRY message. The ORF message contains the patient data and vital signs measurements.
ACK	There are two types of Acknowledgement messages: • Negative ACK: If the QRY is invalid or if there is no data to export, the monitor sends a Negative ACK to the server. If the server finds an error in the exported record, it sends a Negative ACK to the monitor. • Positive ACK: The server sends this message back to the Efficia CM Series monitor to acknowledge that the ORF message was received and validated.

Note — For a complete description of HL7 message syntax, see the *Health Level Seven Standard Version 2.4* available from Health Level Seven International (www.hl7.org).

Notation Conventions

The HL7 messages use the following notation conventions:

- Single ASCII characters are enclosed in single quotes.
- Special characters or non-printing ASCII characters are enclosed in angle brackets, < >. Special characters are the Lower Layer Protocol (LLP) Start Block and End Block characters. Non-printing ASCII characters can be written as abbreviations; for example, ESC for the Escape character. They also may be written as hex values in the form 0xXX, where X is a hexadecimal digit. For example, in Standard ASCII, <ESC> is <0x1B >.
- The HL7 Minimal Lower Layer Protocol (MLLP) is used to mark message boundaries.

Block Format

HL7 messages are enclosed by special characters to form a block. The format is as follows:

```
<SB>dddd<EB><CR>
```

where:

<SB> = Start Block character (1 byte). The ASCII vertical tab character <VT>, for example, <0x0B>. Do not confuse this with the ASCII characters SOH or STX.

dddd = Data (variable number of bytes). This is the HL7 data content of the block. The data can contain any ASCII characters and the carriage return character <CR>, as a delimiter of individual message segments. UTF-8 encoding is used for languages that cannot use ASCII characters.

<EB> = End Block character (1 byte). The ASCII field separator character <FS>, for example, <0x1C>. Do not confuse this with the ASCII characters ETX or EOT.

<CR> = Carriage Return (1 byte). The ASCII carriage return character, for example, <0x0D>.

Delimiter Characters

The following characters are used to construct a message:

Character	Meaning
	Field Separator. Separates two adjacent data fields within a segment.
^	Component Separator. Separates adjacent components of data fields where allowed.
&	Subcomponent Separator. Separates adjacent subcomponents of data fields where allowed.
~	Repetition Separator. Separates multiple occurrences of a field where allowed.
\	Escape Character. Escape character for use with any field.

Segment Data Fields

Each message segment description contains the following data fields:

Data Field	Description
Seq	Sequence number of the field within the segment. Since an Efficia monitor does not use all possible fields in the HL7 standard, the sequence numbers are not always consecutive. This document shows only the fields that are included in the messages.
Field	Descriptive name for the field.
Description	Data field description.
DT	Data type of the field. Some common data types are: • ST = String • NM = Numeric • TS = Time stamp • CE = Coded element
Len	Maximum length of the field (for primitive types only).
Usage	Indicates whether the field is required or conditional in the segment. The abbreviations are: • R = Required element • RE = Required, but can be empty • C = Conditional, based on predicate evaluation • CE = Conditional, but can be empty
Min	Minimum number of occurrences.
Max	Maximum number of occurrences.
Table	Number of the HL7 table containing valid values for the field.
Sample Value	An example of the data field contents.

Query (QRY) Message Syntax

The server sends a Query (QRY) message to the monitor to request current patient data. The QRY message triggers the event R02 (query for results of observation).

QRY Message Segments

The following table describes the segments within the QRY message:

Segment	Description	Usage	Min	Max
MSH	Message Header	R	1	1
QRD	Query Definition	R	1	1

Note — Each message segment consists of a variable number of lines, ending with a carriage return character. In the examples, long lines are broken for display purposes, and the carriage return character is indicated by <CR>.

Examples:

The following example shows a QRY message sent by the server to request all parameters in the numeric panes on the monitor.

```
MSH|^~\&|||||20140510124500||QRY^R02|EXAMPLE123|P|2.4<CR>
QRD|20120510124500|R|I|QUERY123|||||STA<CR>
```

Message Header (MSH) Segment

Every QRY message begins with the MSH segment, which contains message envelope information, such as the timestamp, message type, and some specifics of the message syntax. The MSH segment does not contain clinical information.

Example:

This MSH segment shows a QRY message being sent from the server on May 10, 2014, at 12:45:00 pm.

```
MSH|^~\&|||||20140510124500||QRY^R02|EXAMPLE123|P|2.4<CR>
```

The following table describes each field in the MSH segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Field Separator	The character that separates the segment ID (MSH) from the first data field (Encoding Characters) in the segment.	ST	1	R	1	1		

Query (QRY) Message Syntax

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
2	Encoding Characters	The four separator characters for the message in this order: component separator (^), repetition separator (~), escape character (\), subcomponent separator (&).	ST	4	R	1	1		^~\&
7	Date/Time of Message	The date and time that the monitor created the message.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDH HMMSS.	NM	20	R	1	1		20140510124500
9	Message Type	The server uses this field to know which data segments to recognize. Contains the message type and trigger event code.	CM_MSG	15	R	1	1	0076	QRY^R02
9.1	message type	Query (QRY) message.	ID	3	R	1	1	0076	QRY
9.2	trigger event	Query for results of observation: R02 .	ID	3	R	1	1	0003	R02
10	Message Control ID	The unique identifier for this message instance. The receiving system echoes this ID back to the sending system in the MSA segment.	ST	20	R	1	1		EXAMPLE123

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
11	Processing ID	Indicates whether to process the message as part of a production (P), training (T), or debugging (D) system.	PT	3	R	1	1		
11.1	processing ID	All results from the monitor are production (P) messages.	ID	3	R	1	1	0103	P
12	Version ID	The HL7 version that the server matches to its own version to be sure the message will be interpreted correctly. HL7 is backwards compatible; any previous version is compatible with this specification.	VID	8	R	1	1	0104	
12.1	version ID	Version 2.4 is currently supported.	ID	3	R	1	1	0104	2.4

Query Definition (QRD) Segment

The QRD segment specifies the date and time of the query, the format code, and the priority. The QRD segment is sent with the QRY message and then returned in the ORF message.

Examples:

This QRD segment shows the query message being sent to the Efficia CM Series monitor on May 10, 2014, at 12:45:00 pm. The receiver requests the status of all parameters in the numeric panes on the monitor.

```
QRD|20140510124500|R|I|QUERY123||||STA<CR>
```

The following table describes each field in the QRD segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Query Date/Time	The query date and time. This field is ignored by the monitor, but the given value is sent to the server in the ORF message. Note — <i>The server should supply this value in order to be HL7 compliant.</i>	TS	26	R	1	1		20140510124500
2	Query format code	Must always be R for record-oriented, The monitor does not check the format code.	ID	1	R	1	1		R
3	Query Priority	Must always be Immediate.(I).	ID	1	R	1	1		I
4	Query ID	A unique identifier. This field must not be empty. The Query ID is not processed, but is sent to the server in the ORF message. Note — <i>The monitor does not check whether query IDs are unique.</i>	ST	10	R	1	1		QUERY123
7	Quantity Limited Request	Always empty for NO LIMIT. This field is ignored.	CQ	10		0	0		
9	What subject filter	Filter must be the same in query and response. Valid values are: • STA = Status, screen display	CE	60	R	1	1		STA

Observation Reporting (ORF) Message Syntax

The Observation Reporting (ORF) message is used to report patient data as a reply to a query. It may contain any of the following information:

- Patient information
 - Medical Record Number (MRN) or other ID used as the patient identifier
 - First Name
 - Middle Name
 - Last Name
 - Gender
 - Age
 - Height
 - Weight
- Monitor Name
- Location ID (also called Bed ID)
- Transaction ID (also called Visit Number, Account Number, Finance Number (FIN))
- Operator ID
- Department
- Admit Time
- All parameters currently available on the monitor

Note — *The person configuring the monitor specifies which patient identifiers are required or optional, and selects the primary identifier. For information about configuring patient IDs, see the Efficia CM Series Patient Monitors Series Installation and Configuration Guide.*

ORF Message Segments

The following table describes the segments within the ORF message:

Segment	Description	Usage	Min	Max
MSH	Message Header	R	1	1
MSA	Message Acknowledgement	R	1	1
QRD	Query Definition (from QRY message)	R	1	1
PID	Patient Identification	R	1	1
PV1	Patient Visit	R	1	1
ORC	Common Order	R	1	1
OBR	Observation Request	R	1	1

Segment	Description	Usage	Min	Max
OBX	Observation/Result	R	1	no max

Note — Each message segment consists of a variable number of lines, ending with a carriage return character. In the examples, long lines are broken for display purposes, and the carriage return character is indicated by <CR>.

Examples:

The following example shows a ORF message sent by an Efficia CM Series monitor to report the status of all parameters in the numeric panes.

```
MSH|^~\&|^EfficiaCM|||20140510124500||ORF^R04|MONITORABC000000002D|P|
2.4|||||||2.16.840.1.113883.9.2.1<CR>
MSA|AA|EXAMPLE123<CR>
QRD|20140510124500|R|I|QUERY123||||STA<CR>
PID|||MRN1234||JONES^MARY^A<CR>
PV1|||^LOCATION34|||||||A|TRANSACTION1234<CR>
ORC|NW|||||||MONITORABC<CR>
OBR|||MONITOR||20140510124500||OPERATOR12<CR>
OBX||NM|0002-4bb8^SpO2^MDIL||98|0004-0220^%MDIL||||F<CR>
OBX||NM|0002-4a05^NBPs^MDIL||122|0004-
0f20^mmHg^MDIL||||F||20140510124430<CR>
OBX||NM|0002-4a06^NBPd^MDIL||85|0004-
0f20^mmHg^MDIL||||F||20140510124430<CR>
OBX||NM|0002-4a07^NBPm^MDIL||98|0004-0f20^mmHg^MDIL||||F<CR>
OBX||NM|0002-480a^Pulse^MDIL||80|0004-0aa0^bpm^MDIL||||F||||
|0002-4bb8^SpO2^MDIL<CR>
OBX||NM|0002-5000^Resp^MDIL||20|0004-0ae0^rpm^MDIL||||F<CR>
OBX||NM|0002-f0c7^T1^MDIL||37.4|0004-1710^C^MDIL||||F<CR>
```

Message Header (MSH) Segment

Every ORF message begins with the MSH segment, which contains message envelope information, such as the source, destination, and some specifics of the message syntax. The MSH segment does not contain clinical information.

Example:

This MSH segment shows an ORF message being sent from the Efficia CM Series series monitor on May 10, 2014, at 12:45:00 pm.

```
MSH|^~\&|^EfficiaCM|||20140510124500||ORF^R04|MONITORABC000000002D|P|
2.4|||||||2.16.840.1.113883.9.2.1<CR>
```

The following table describes each field in the MSH segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Field Separator	The character that separates the segment ID (MSH) from the first data field (Encoding Characters) in the segment.	ST	1	R	1	1		
2	Encoding Characters	The four separator characters for the message in this order: component separator (^), repetition separator (~), escape character (\), subcomponent separator (&).	ST	4	R	1	1		^~\&
3	Sending Application	Generic identifier for the Efficia monitor.	HD	180	R	1	1	0361	^EfficiaCM
3.2	universal ID	The identifier for all Efficia monitors.	ST	20	R	1	1		EfficiaCM
7	Date/Time of Message	The date and time that the monitor created the message.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDHH MMSS.	NM	20	R	1	1		20140510124500
9	Message Type	The server uses this field to know which data segments to recognize. Contains the message type and trigger event for the message.	CM_MSG	15	R	1	1	0076	ORF^R04
9.1	message type	Observation Reporting (ORF) message.	ID	3	R	1	1	0076	ORF
9.2	trigger event	Response to query; transmission of requested observation: R04 .	ID	3	R	1	1	0003	R04

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
10	Message Control ID	The unique identifier for this message instance. The identifier is a combination of the 10-character device serial number and a 10-character message counter.	ST	20	R	1	1		MONITORABC000000002D
11	Processing ID	Indicates whether to process the message as part of a production (P), training (T), or debugging (D) system.	PT	3	R	1	1		
11.1	processing ID	All results from the monitor are production (P) messages.	ID	3	R	1	1	0103	P
12	Version ID	The HL7 version that the server matches to its own version to be sure the message will be interpreted correctly. HL7 is backwards compatible; any previous version is compatible with this specification.	VID	8	R	1	1	0104	
12.1	version ID	Version 2.4 is currently supported.	ID	3	R	1	1	0104	2.4
21	Conformance Statement ID	The HL7 registry object identifier (OID) corresponding to this message profile.	ID	30	R	1	1	0449	2.16.840.1.113883.9.2.1

Message Acknowledgement (MSA) Segment

The MSA segment in the ORF message contains an acknowledgement to the server's QRY message.

Example:

This MSA segment shows that the monitor acknowledges the QRY message from the server.

```
MSA | AA | EXAMPLE123<CR>
```

The following table describes each field in the MSA segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Acknowledgment Code	Contains the acknowledgment code. AA = Application Accept (positive acknowledgment).	ID	2	R	1	1		AA
2	Message Control ID	Contains the message control ID of the QRY message sent by the server.	ST	20	R	1	1		EXAMPLE123

Query Definition (QRD) Segment

The QRD segment in the ORF message is the same QRD segment that was sent in the QRY message.

```
QRD|20140510124500|R|I|QUERY123|||||RES<CR>
```

This QRD segment shows the query message being sent to the monitor on May 10, 2014, at 12:45:00 pm. The receiver requests the status of all parameters in the numeric panes on the monitor.

```
QRD|20140510124500|R|I|QUERY123|||||STA<CR>
```

The following table describes each field in the QRD segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Query Date/Time	Contains the date and time from the incoming QRY message; this is used for the ORF response message.	TS	26	R	1	1		20140510124500
2	Query format code	The format of the response. Must always be record-oriented (R).	ID	1	R	1	1		R
3	Query Priority	The timeframe in which the response is expected. Must always be Immediate (I).	ID	1	R	1	1		I
4	Query ID	A unique identifier for the query.	ST	10	R	1	1		QUERY123

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
7	Quantity Limited Request	The maximum length of the response that can be accepted by the requesting system. Always empty (NO LIMIT).	CQ	10		0	0		
9	What subject filter	Describes the kind of information required to satisfy the request. Valid values are: STA = Status	CE	60	R	1	1		STA

Patient Identification (PID) Segment

The PID segment in the ORF message identifies the subject of the measurements.

Example:

This PID segment identifies the patient as Mary A. Jones. Her medical record number is MRN1234. Her birth date is September 11, 1991.

```
PID|||MRN1234||JONES^MARY^A||19910900|F<CR>
```

The following table describes each field in the PID segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
3	Patient Identifier List	Contains the patient ID entered in the device.	CX	250	RE	0	1		
3.1	ID	Patient identifier. The medical record number (MRN) or another unique identifier used across the enterprise.	ST	20	RE	0	1		MRN1234
5	Patient Name	The patient's name.	XP	45	RE	0	1		JONES^MARY^A
5.1	family name	The patient's family name.	FN	15	RE	0	1		
5.1.1	surname	The patient's last name.	ST	15	RE	0	1		JONES
5.2	given name	The patient's first name.	ST	15	RE	0	1		MARY

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
5.3	middle initial or name	The patient's middle name or initial.	ST	15	RE	0	1		A
7	Date of Birth	The patient's date of birth in the format YYYYMMDD	NM	8	RE	0	1		19910911
8	Gender	The patient's gender F – Female M – Male Blank – Unknown or not entered by the user	IS	2	RE	0	1		F

Patient Visit (PV1) Segment

The PV1 segment in the ORF message identifies the location ID, patient type and transaction (visit) ID.

Example:

This PV1 segment shows that the patient's bed is at LOCATION34. The patient is an adult and the visit ID is TRANSACTION1234.

```
PV1|||^^LOCATION34|||||||A|TRANSACTION1234<CR>
```

The following table describes each field in the PV1 segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
3	Assigned Patient Location	The patient's assigned location.	PL	80	RE	0	1		^^LOCATION34
3.3	bed	Location ID for the bed.	IS	12	RE	0	1		LOCATION34
18	Patient Type	Identifies the patient type. The abbreviations are: • A = Adult • P = Pediatric • N = Neonatal	IS	2	R	1	1	0018	A
19	Visit Number	Unique number assigned to the patient visit.	CX	250	RE	0	1		
19.1	ID	Transaction or visit ID.	ST	20	RE	0	1		TRANSACTION1234
39	Department	Department to which patient is admitted	ST	20	RF	0	1		

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
44	Admit Time	Time when patient is admitted in the format YYYYMMDDHH MMSS	NM	14	RE	0	1		

Common Order (ORC) Segment

The ORC segment in the ORF message transmits the monitor name.

Example:

This ORC segment contains the monitor name MONITORABC:

```
ORC|NW| | | | | | | | | | | | | |MONITORABC<CR>
```

The following table describes each field in the ORC segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Order Control	The function of the order segment. For example, NW = new order.	ID	2	R	1	1	0119	NW
18	Entering Device	Identifies the physical device (monitor) used to enter the vital signs measurements.	CE	250	R	1	1		
18.1	ID	Monitor name.	ST	10	R	1	1		MONITORABC

Observation Request (OBR) Segment

The OBR segment in the ORF message identifies the type of data being exported, the date and time that the measurements occurred, and the ID of the person taking the measurements. If the monitor is sending trend data, the data type in the OBR segment is MONITOR.

Examples:

This OBR segment identifies the type of results as trend data. The measurements occurred at 12:45:00 pm on May 10, 2014 and were collected by OPERATOR12.

```
OBR|||MONITOR|||20140510124500|||OPERATOR12<CR>
```

The following table describes each field in the OBR segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
4	Universal Service Identifier	The identifier code for the requested measurement.	CE	250	R	1	1		

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
4.1	identifier	Identifies the type of result. Valid values are: • MONITOR	ST	20	R	1	1		MONITOR
7	Observation Date/Time	Date and time of the measurement.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDHHMMSS.	NM	20	R	1	1		20140510124500
10	Collector Identifier	Identifies the person collecting the measurements.	XCN	20	RE	0	1		
10.1	ID number	The ID of the person collecting the measurements.	ST	12	RE	0	1		OPERATOR12

Observation/Result (OBX) Segment

Each OBX segment in the ORF message reports a single measurement. There can be multiple OBX segments in an ORF message.

Examples:

These OBX segments contain the following measurements: SpO₂ (98%); systolic NBP (122 mmHg); diastolic NBP (85 mmHg); mean NBP (98 mmHg); Pulse Rate (80 bpm) derived from SpO₂; ECG respiration (20 rpm); Predictive temperature (37.4°C). All of the results are final.

```
OBX||NM|0002-4bb8^SpO2^MDIL||98|0004-0220^% ^MDIL||||F<CR>
OBX||NM|0002-4a05^NBPs^MDIL||122|0004-
0f20^mmHg^MDIL||||F|||20120510123500<CR>
OBX||NM|0002-4a06^NBPd^MDIL||85|0004-
0f20^mmHg^MDIL||||F|||20120510123500<CR>
OBX||NM|0002-4a07^NBPm^MDIL||98|0004-
0f20^mmHg^MDIL||||F|||20120510123500<CR>
OBX||NM|0002-480a^Pulse^MDIL||80|0004-0aa0^bpm^MDIL||||F|||||
|0002-4bb8^SpO2^MDIL<CR>
OBX||NM|0002-5000^Resp^MDIL||20|0004-0ae0^rpm^MDIL||||F<CR>
OBX||NM|0002*f0c7^Temp^MDIL||37.4|0004-1710^C^MDIL||||F|||||0002-
f110^Oral^MDIL<CR>
```

In these OBX segments, the empty Observation Value field (OBX-5) indicates that there is no NBP measurement on the monitor screen:

```
OBX||NM|0002-4a05^NBPs^MDIL|||0004-0f20^mmHg^MDIL||||F<CR>
OBX||NM|0002-4a06^NBPd^MDIL|||0004-0f20^mmHg^MDIL||||F<CR>
OBX||NM|0002-4a07^NBPm^MDIL|||0004-0f20^mmHg^MDIL||||F<CR>
```

In this OBX segment, the **X** in the Observation Result Status field (OBX-11) indicates an invalid SpO₂ measurement, which is displayed as a question mark (?) on the monitor screen.

```
OBX||NM|0002-4bb8^SpO2^MDIL|||0004-0220^%^MDIL||||X<CR>
```

In this OBX segment, the empty Observation Value field (OBX-5) indicates that HR pane on the monitor screen contains a question mark (?). The **X** in the Observation Result Status field (OBX-11) indicates an invalid measurement:

```
OBX||NM|0002-4182^HR^MDIL|||0004-0aa0^bpm^MDIL||||X<CR>
```

The following table describes each field in the OBX segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
2	Value Type	Format of the measurement in the Observation Value field. The format is usually NM (numeric).	ID	2	R	1	1	0125	NM
3	Observation Identifier	Contains a unique identifier for the measurement. The format is: <identifier>^ <text>^ <name of coding system>	CE	250	R	1	1		0002-4182^HR^MDIL
3.1	identifier	Code that uniquely identifies the measurement. The identifier is an MDIL code. For more information, see “MDIL Coding” on page 3-27.	ST	20	R	1	1		0002-4182
3.2	text	Name or description of the coded measurement.	ST	20	R	1	1		HR
3.3	name of coding system	For measurements, the coding system is MDIL. For more information, see “MDIL Coding” on page 3-27.	IS	20	RE	0	1	0396	MDIL
5	Observation Value	Contains the measurement value recorded on the monitor. If there is no reading, or a question mark (?) appears on the monitor, this field is empty.	NM	100	RE	0	1		80

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
6	Units	Contains the units of measure for the Observation Value. The format is: <identifier>^ <unit of measure>^ <name of coding system>	CE	250	R	1	1		0004-0aa0^bpm^MDIL
6.1	identifier	Code that uniquely identifies the unit of measurement. The identifier is an MDIL code. For more information, see “MDIL Coding” on page 3-27.	ST	20	R	1	1		0004-0aa0
6.2	text	The name or description of the unit of measure.	ST	20	R	1	1		bpm
6.3	name of coding system	For measurements, the coding system is MDIL. For more information, see “MDIL Coding” on page 3-27.	IS	20	RE	0	1	0396	MDIL
8	Abnormal Flags	Indicates the normalcy status of the result. Valid values are: • Empty = Normal • L = Below normal • H = Above normal • A = Other abnormal	ID	1	RE	0	1	0078	H
11	Observation Result Status	Contains the result status. Valid values are: • F = Final results • X = Results cannot be obtained	ID	1	R	1	1	0085	F

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
14	Observation Date/Time	Timestamp of the following measurements: - NBP - Pulse Rate derived from NBP Timestamp is in the format YYYYMMDDHH MMSS	NM	20	RE	0	1		20140510123500
17	Observation Method	The method used to obtain the result.	CE	250	RE	0	1		0002-4bb8^SpO2^MDIL
17.1	identifier	The code that corresponds to the Heart Rate source, Pulse Rate source, or temperature method. By default, these measurements use MDIL codes.	ST	20	RE	0	1		0002-4bb8
17.2	text	Text describing the measurement method. Valid methods are: Pulse Rate source: SpO2 or NBP	ST	20	RE	0	1		SpO2
17.3	name of coding system	For Heart Rate source, Pulse Rate source, and temperature method, the default coding system is MDIL. For more information, see "MDIL Coding" on page 3-27.	IS	20	RE	0	1	0396	MDIL

Acknowledgement (ACK) Message Syntax

There are two types of Acknowledgement (ACK) messages:

- **Negative ACK:** If the QRY is invalid or if there is no patient record to export, the monitor sends a Negative ACK to the server.
- **Positive ACK:** This message is sent by the server back to the Efficia CM Series monitor. It acknowledges that the message was received and validated.

Both of these ACK messages are described in the following sections.

Negative ACK Message Syntax

The Negative ACK message is sent from the monitor to the server if the QRY from the server is invalid or if there is no patient record to export.

Negative ACK Message Segments

The following table describes the segments within the Negative ACK message.

Segment	Description	Usage	Min	Max
MSH	Message Header	R	1	1
MSA	Message Acknowledgement	R	1	1

Note — Each message segment consists of a variable number of lines, each ending with a carriage return character. In the examples, long lines are broken for display purposes, and the carriage return character is indicated by <CR>.

Examples:

The following Negative ACK message from the monitor to the server indicates that the query is invalid.

```
MSH|^~\&|^EfficiaCM|||20140510124500||ACK^|MONITORABC000000002D|P|2.4
|||||
2.16.840.1.113883.9.2.1<CR>
MSA|AE|<CR>
```

The following Negative ACK message from the monitor to the server indicates that there is no patient data to export.

```
MSH|^~\&|^EfficiaCM|||20140510124500||ACK^|MONITORABC000000002D|P|2.4
|||||
2.16.840.1.113883.9.2.1<CR>
MSA|AE|EXAMPLE123<CR>
```

The following Negative ACK message from the server to the monitor indicates that there is a problem with the exported data.

```
MSH|^~\&|||20140510124500||ACK^|EXAMPLE123|P|2.4<CR>
MSA|AE|MONITORABC000000002D|Incorrect Patient ID<CR>
```

Message Header (MSH) Segment

The MSH segment is used primarily as a message envelope.

Example:

This MSH segment shows a Negative ACK message being sent from the monitor to the server on May 10, 2014 at 12:45:00 pm.

```
MSH|^~\&|^EfficiaCM|||20140510124500||ACK^|MONITORABC000000002D|P|2.4
|||||2.16.840.1.113883.9.2.1<CR>
```

The following table describes each field in the MSH segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Field Separator	The character that separates the segment ID (MSH) from the first data field (Encoding Characters) in the segment.	ST	1	R	1	1		
2	Encoding Characters	The four separator characters for the message in this order: component separator (^), repetition separator (~), escape character (\), subcomponent separator (&).	ST	4	R	1	1		^~\&
3	Sending Application	Generic identifier for the Efficia monitor.	HD	180	R	1	1	0361	^EfficiaCM
3.2	universal ID	The identifier for all Efficia monitors.	ST	20	R	1	1		EfficiaCM
7	Date/Time of Message	The date and time that the monitor created the message.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDHH MMSS.	NM	20	R	1	1		20140510124500
9	Message Type	Contains the message type.	CM_M SG	15	R	1	1	0076	ACK^
9.1	message type	Acknowledgement (ACK) message.	ID	3	R	1	1	0076	ACK
10	Message Control ID	The unique identifier for this message instance.	ST	20	R	1	1		MONITORABC000000002D
11	Processing ID	Indicates whether to process the message as part of a production (P), training (T), or debugging (D) system.	PT	3	R	1	1		

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
11.1	processing ID	All results from the monitor are Production (P) messages.	ID	3	R	1	1	0103	P
12	Version ID	The HL7 version that the server matches to its own version to be sure the message will be interpreted correctly. HL7 is backwards compatible; any previous version is compatible with this specification.	VID	8	R	1	1	0104	
12.1	version ID	Version 2.4 is currently supported.	ID	3	R	1	1	0104	2.4
21	Conformance Statement ID	The HL7 registry object identifier (OID) corresponding to this message profile.	ID	30	R	1	1	0449	2.16.840.1.113883.9.2.1

Message Acknowledgement (MSA) Segment

The MSA segment in a Negative ACK message contains the response code and the initiating message identifier.

Examples:

This MSA segment shows that the server sent an invalid query.

```
MSA|AE|<CR>
```

This MSA segment shows that there is no patient record to export.

```
MSA|AE|EXAMPLE123<CR>
```

This MSA segment from the server shows that there was a problem with the patient data from the monitor.

```
MSA|AE|MONITORABC000000002D|Incorrect Patient ID<CR>
```

The following table describes each field in the MSA segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Acknowledgment Code	Contains the acknowledgment code. AE = Acknowledgment Error.	ID	2	R	1	1	0008	AE

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
2	Message Control ID	Contains the message control ID of the initiating message (QRY).	ST	20	RE	1	1		MONITORABC000000002D
3	Text Message	If the exported record contains an error, the server can send a text message describing the error.	ST	80	RE	0	1		Incorrect Patient ID

Positive ACK Message Syntax

The Positive ACK message is sent by the server back to the Efficia CM Series monitor. It acknowledges that the message was received and validated.

Positive ACK Message Segments

The following table describes the segments within the ACK message.

Segment	Description	Usage	Min	Max
MSH	Message Header	R	1	1
MSA	Message Acknowledgement	R	1	1

The following syntax message information represents the profile for the receiver. Any message segment not supported is ignored. RE elements (**required, but can be empty**) will be used, if valid, but ignored otherwise.

Note — Each message segment consists of a variable number of lines, each ending with a carriage return character. In the examples, long lines are broken for display purposes, and the carriage return character is indicated by <CR>.

Example:

The following example shows a Positive ACK message sent by the server back to the monitor to acknowledge that the message was received and validated.

```
MSH|^~\&||||20140510124500||ACK^^ACK_ALL|EXAMPLE123|P|2.4<CR>
MSA|AA|MONITORABC000000002D<CR>
```

Message Header (MSH) Segment

The MSH segment is used primarily as a message envelope.

Example:

This MSH segment shows a Positive ACK message being sent from the server to the monitor on May 10, 2014 at 12:45:00 pm.

```
MSH|^~\&||||20140510124500||ACK^^ACK_ALL|EXAMPLE123|P|2.4<CR>
```

The following table describes each field in the MSH segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Field Separator	The character that separates the segment ID (MSH) from the first data field (Encoding Characters) in the segment.	ST	1	R	1	1		
2	Encoding Characters	The four separator characters for the message in this order: component separator (^), repetition separator (~), escape character (\), subcomponent separator (&).	ST	4	R	1	1		^~\&
7	Date/Time Of Message	The date and time of the message transmission.	TS	26	R	1	1		
7.1	Date/Time	Timestamp in the format YYYYMMDDHH MMSS.	NM	20	R	1	1		20140510124500
9	Message Type	Contains the message type and message structure.	CM_M SG	15	R	1	1	0076	ACK^^ACK_ALL
9.1	message type	Acknowledgment (ACK) message.	ID	3	R	1	1	0076	ACK
9.3	message structure		ID	7	R	1	1	0354	ACK_ALL
10	Message Control ID	The unique identifier for this message instance.	ST	20	R	1	1		EXAMPLE123

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
11	Processing ID	Indicates whether to process the message as part of a production (P), training (T), or debugging (D) system.	PT	3	R	1	1		
11.1	processing ID	All results from the monitor are Production (P) messages.	ID	3	R	1	1	0103	P
12	Version ID	The HL7 version that the server matches to its own version to be sure the message will be interpreted correctly. HL7 is backwards compatible; any previous version is compatible with this specification.	VID	8	R	1	1	0104	
12.1	version ID	Version 2.4 is currently supported.	ID	3	R	1	1	0104	2.4

Message Acknowledgement (MSA) Segment

The MSA message contains the response code and the initiating message identifier.

Example:

This MSA segment shows that the server successfully received and validated the ORF message from the monitor.

```
MSA|AA|MONITORABC000000002D<CR>
```

The following table describes each field in the MSA segment.

Seq	Field	Description	DT	Len	Usage	Min	Max	Table	Sample Value
1	Acknowledgment Code	Contains the acknowledgment code. The following codes are supported: • AA = Accept • AR = Reject • AE = Error	ID	2	R	1	1	0008	AA
2	Message Control ID	Contains the message control ID of the initiating message (ORF).	ST	20	R	1	1		MONITORABC000000002D

MDIL Coding

The Efficia CM Series monitor uses numeric codes from the Medical Device Interface Language (MDIL) standard to uniquely identify parameters and unit sources.

Parameters and Unit Sources

The MDIL identifiers use the following format:

<Partition>--<TermCode>

The values for the Partition and TermCode are hexadecimal values between 0x0000 and 0xffff, which are separated in the output with a “-” character.

Partition may be one of the following values:

- 0x0002 - SCADA partition
- 0x0401 - EMFC partition
- 0x0004 - Dimension partition (for units)

Note — *The label strings are provided in English only. The strings are not translated into other languages. Philips recommends that for parameter identification, you use the numeric parameter identifiers instead of parameter labels.*

The following table lists the MDIL codes for Efficia CM Series monitor parameters. Only parameters that are available on your monitor are exported.

MDIL Code	Parameter Label	Description
0002-4bb8	SpO2	Oxygen saturation of arterial blood
0002-4a05	NBPs	Non-invasive blood pressure (systolic)
0002-4a06	NBPd	Non-invasive blood pressure (diastolic)
0002-4a07	NBPm	Non-invasive blood pressure (mean)
0002-4182	HR	Heart Rate
0002-480a	Pulse	Pulse Rate
0002-4a15	ABPs	Arterial blood pressure (systolic)
0002-4a16	ABPd	Arterial blood pressure (diastolic)
0002-4a17	ABPm	Arterial blood pressure (MAP)
0002-4a1d	PAPs	Pulmonary artery pressure (systolic)
0002-4a1e	PAPd	Pulmonary artery pressure (diastolic)
0002-4a1f	PAPm	Pulmonary artery pressure (MAP)
0002-4a47	CVPm	Central venous pressure (mean)
0002-50b0	etCO2	End tidal CO ₂ concentration

MDIL Code	Parameter Label	Description
0002-50ba	imCO2	Inspired minimum CO ₂
0002-5012	awRR	Airway respiration rate
0002-5000	Resp	ECG Respiration
0002-0301	ST-I	ST of ECG Lead I
0002-0302	ST-II	ST of ECG Lead II
0002-033d	ST-III	ST of ECG Lead III
0002-033e	ST-aVR	ST of ECG Lead aVR
0002-033f	ST-aVL	ST of ECG Lead aVL
0002-0340	ST-aVF	ST of ECG Lead aVF
0002-0343	ST-V	ST of ECG Lead V
0002-034b	ST-MCL	ST of ECG Lead MCL
0002-0303	ST-V1	ST of ECG Lead V1
0002-0304	ST-V2	ST of ECG Lead V2
0002-0305	ST-V3	ST of ECG Lead V3
0002-0306	ST-V4	ST of ECG Lead V4
0002-0307	ST-V5	ST of ECG Lead V5
0002-0308	ST-V6	ST of ECG Lead V6
0002-4261	PVC	Premature ventricular contraction
0002-fb62	SpCO	Carboxyhemoglobin saturation
0002-fb67	SpOC	Oxygen content
0002-fbcf	SpHb	Total hemoglobin
0002-fb5f	RRa	Acoustic respiration rate
0002-4bb0	Perf	Perfusion index
0002-4a11	ARTs	Arterial blood pressure (systolic)
0002-4a12	ARTd	Arterial blood pressure (diastolic)
0002-4a13	ARTm	Arterial blood pressure (mean)
0002-4a0d	Aos	Aortic pressure (systolic)
0002-4a0e	Aod	Aortic pressure (diastolic)
0002-4a0f	Aom	Aortic pressure (mean)
0002-4a01	Ps	Non-specific pressure (systolic)
0002-4a02	Pd	Non-specific pressure (diastolic)
0002-4a03	Pm	Non-specific pressure (mean)

MDIL Code	Parameter Label	Description
0002-4a24	PAWP	Pulmonary artery wedge pressure
0002-4a29	UAPs	Umbilical arterial pressure (systolic)
0002-4a2a	UAPd	Umbilical arterial pressure (diastolic)
0002-4a2b	UAPm	Umbilical arterial pressure (mean)
0002-5809	ICPs	Intracranial pressure (systolic)
0002-580a	ICPd	Intracranial pressure (diastolic)
0002-580b	ICPm	Intracranial pressure (mean)
0002-4a31	LAPs	Left atrial pressure (systolic)
0002-4a32	LAPd	Left atrial pressure (diastolic)
0002-4a33	LAPm	Left atrial pressure (mean)
0002-4a35	RAPs	Right atrial pressure (systolic)
0002-4a36	RAPd	Right atrial pressure (diastolic)
0002-4a37	RAPm	Right atrial pressure (mean)
0002-4a49	UVPs	Umbilical venous pressure (systolic)
0002-4a4a	UVPd	Umbilical venous pressure (diastolic)
0002-4a4b	UVPm	Umbilical venous pressure (mean)
0002-f0c7	T1	Temperature #1
0002-f0c8	T2	Temperature #2
0002-f0c9	T3	Temperature #3
0002-f0ca	T4	Temperature #4
0002-4b04	C.O.	Cardiac output
0002-e014	Tblood	Blood temperature
0002-f093	Weight	Patient weight
0002-f094	Height	Patient height

The following table lists the MDIL codes for the measurements of the plug-in modules connected to the Efficia Rack. Only those parameters that are available on your monitor are exported.

MDIL Code	Parameter Label	Description
0002-f04e	BIS	Bispectral Index
0002-593c	EMG	Electromyography
0002-5988	SEF	Spectral Edge Frequency
0002-f04d	SQI	BIS Signal Quality Index

MDIL Code	Parameter Label	Description
0002-f04a	SR	EEG Suppression Ratio
0002-59b8	TP	Total Power
0002-59ac	BC	Burst Count
0002-f04e	BIS L	Left Bispectral Index
0002-593c	EMG L	Left Electromyography
0002-5988	SEF L	Left Spectral Edge Frequency
0002-f04d	SQI L	Left BIS Signal Quality Index
0002-f04a	SR L	Left EEG Suppression Ratio
0002-59b8	TP L	Left Total Power
0002-59ac	BC L	Left Burst Count
0002-f04e	BIS R	Right Bispectral Index
0002-593c	EMG R	Right Electromyography
0002-5988	SEF R	Right Spectral Edge Frequency
0002-f04d	SQI R	Right BIS Signal Quality Index
0002-f04a	SR R	Right EEG Suppression Ratio
0002-59b8	TP R	Right Total Power
0002-59ac	BC R	Right Burst Count
0002-538c	etAGT	Gas Analyzer Agent - End Tidal
0002-5390	inAGT	Gas Analyzer Agent - inspired
0002-f81e	etAGTs	End Tidal Anesthetic Agent - secondary agent
0002-f81f	inAGTs	Inspired Gas Analyzer Agent - secondary agent
0002-522c	etN2O	Gas Analyzer End-Tidal Nitrous Oxide
0002-5280	inN2O	Gas Analyzer Nitrous Oxide - inspired
0002-5378	etO2	End-Tidal Oxygen Concentration
0002-5284	inO2	Oxygen Concentration - inspired
0002-5214	etDES	Desflurane Concentration - End Tidal
0002-5218	etENF	Enflurane Concentration - End Tidal
0002-521c	etHAL	Halothane Concentration - End Tidal
0002-5220	etSEV	Gas Analyzer Sevoflurane - End Tidal
0002-5224	etISO	Gas Analyzer Isoflurane - End Tidal
0002-5268	inDES	Desflurane Concentration - inspired
0002-526c	inENF	Enflurane Concentration - inspired

MDIL Code	Parameter Label	Description
0002-5270	inHAL	Halothane Concentration - inspired
0002-5274	inSEV	Sevoflurane Concentration - inspired
0002-5278	inISO	Gas Analyzer Isoflurane - inspired
0402-f921	sVMode	Setting: Ventilation mode
0002-5012	awRR	Airway Respiration Rate
0002-514c	MVexp	Expired Minute Volume
0002-5109	PIP	Positive Inspiratory Pressure
0002-510b	MnAwP	Mean Airway Pressure
0002-50dd	PIF	Peak Inspiratory Flow
0002-f8a1	ExpTi	Expiratory Time
0002-513c	TV	Tidal Volume
0002-51a8	PEEP	Positive End-Expiratory Pressure
0002-f8a3	InsTi	Spontaneous Inspiration Time
0002-508c	Cdyn	Dynamic Lung Compliance
0002-7498	FiO2	Fractional Inspired Oxygen
0002-f819	RSBI	Rapid Shallow Breathing Index
0002-50e0	VCO2	CO2 Production
0002-fa06	WOBv	Work of Breathing
0002-50b0	etCO2	End-Tidal CO2 Concentration
0002-50e8	Pplat	Plateau Pressure
0002-51cc	MMV	Mandatory Minute Volume
0002-f091	SpMV	Spontaneous Minute Volume
0002-f816	ETC	Time constant
0002-f815	SpAWRR	Spontaneous Airway Respiration Rate
0002-f897	TOFrat	Train of Four Ratio
0002-f8ab	TOFcnt	Train of Four (TOF) count - number of TOF responses
0002-f8ac	Twitch	Twitch height of the 1 Hz/0.1 Hz stimulation response
0002-f86b	DBSrat	Double Burst Ratio
0002-f88b	PTC	Post Tetanic Count stimulation

The following table lists the MDIL codes for units of measurement.

MDIL Code	Parameter Label	Description
0002-f099	MAC	Minimum Alveolar Concentration
0004-0220	%	Percent
0004-0f20	mmHg	Millimeters of mercury
0004-0f03	kPa	Kilopascals
0004-0aa0	bpm	Beats per minute
0004-17a0	C	Celsius
0004-1140	F	Fahrenheit
0004-0ae0	rpm	Respirations per minute
0004-0000	No label	No unit of measure
0004-09e0	/min	per minute
0004-0512	mm	Millimeter
0004-0f40	cmH ₂ O	Centimeter of water
0004-0892	msec	Millisecond
0004-0840	g/dl	Grams per deciliter
0004-1272	mmol/l	Millimoles per liter
0004-1912	ml/dl	Milliliters per deciliter
0004-0c00	l/min	Liters per minute
0004-06c3	kg	Kilogram
0004-06e0	lb	Pound
0004-0511	cm	Centimeter
0004-0560	in	Inch

The following table lists MDIL codes Pulse Rate source.

MDIL Code	Label	Description
0002-4a07	NBPm	Mean NBP Pulse Rate source
0002-4bb8	SpO ₂	SpO ₂ Pulse Rate source

Time Synchronization

This chapter describes how the Efficia CM Series patient monitors synchronize with the clock on the EHR server or HL7 interface server.

Note — *If you have a Simple Network Time Protocol (SNTP) server running in your hospital network, you can configure the monitor to synchronize its clock with that server. For more information, see the Efficia CM Series Patient Monitors Installation and Configuration Guide.*

- If your monitor is configured for LAN data export, the monitor clock can be synchronized with the clock on your EHR system, HL7 interface server, or SNTP server.
- If your monitor is configured for serial data export, the monitor clock can be synchronized with the clock on the computer that receives the serial data from the monitor.

During the data export process, the monitor sends patient data to the receiving system, and then expects an Acknowledgement (ACK) message from the receiving system indicating that the patient data was received and validated. The Message Header (MSH) segment of the ACK message contains a timestamp, shown in bold type in the following example:

```
MSH|^~\&|||||20140501002459||ACK^^ACK_ALL|398098728972|P|2.4
```

Maintaining Synchronization on a LAN System

To keep the clock updated on a wired LAN network, the monitor automatically sends a “blank” Observation Received (ORU) message to the receiving system each time the monitor is turned on and once every hour, as long as the monitor is turned on. This ensures that the monitor clock remains synchronized with the central clock, even during times when the monitor is not actively exporting patient data.

The blank ORU message contains only a Message Header (MSH) segment; it does not contain any patient data, such as PID, PV1, or OBX segments.

Example:

```
MSH|^~\&|^EfficiaCM||||20140510052137||ORU^R01^ORU_R01|USWIFILP022012051005|P|2.4|||||||2.16.840.1.113883.9.2.1
```




© 2019 Koninklijke Philips N.V.

All rights are reserved. Reproduction or transmission in whole or in part, in any form or by any means, electronic, mechanical or otherwise, is prohibited without the prior written consent of the copyright owner.

Printed in USA

453564623581, Revision B

